Exercise J-M

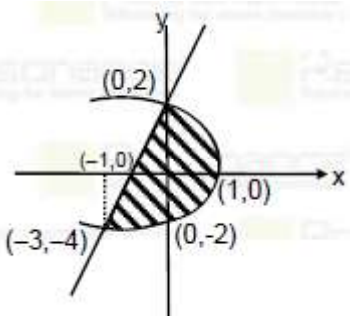
1. The area enclosed by the curves $y^2 + 4x = 4$ and $y - 2x = 2$ is : **[JEE(Main)Jan.2023]**

   वक्रो $y^2 + 4x = 4$ तथा $y - 2x = 2$ से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

   (A) $\frac{25}{3}$ (B) 9 (C) $\frac{23}{3}$ (D) $\frac{22}{3}$

Ans. B

Sol.



Required area $= \int_{-3}^{0} \sqrt{4-4x}dx - \frac{1}{2}(2)(4) + \frac{1}{2}(1)(2) + 2\int_{0}^{1} \sqrt{4-4x}dx$

$= 2\int_{-3}^{0} \sqrt{1-x}dx + 4\int_{0}^{1} \sqrt{1-x}dx - 3$

$= -2 \times \frac{2}{3}\left[(1-x)^{3/2}\right]_{-3}^{0} - 4\frac{2}{3}\left[(1-x)^{3/2}\right] - 3$

$= -\frac{4}{3}(1-8) - \frac{8}{3}(0-1) - 3$

$= \frac{28}{3} + \frac{8}{3} - 3 = 9$

2. Let T and C respectively be the transverse and conjugate axes of the hyperbola $16x^2 - y^2 + 64x + 4y + 44 = 0$. Then the area of the region above the parabola $x^2 = y + 4$, below the transverse axis T and on the right of the conjugate axis C is : **[JEE(Main)Jan.2023]**

   माना अतिपरवलय $16x^2 - y^2 + 64x + 4y + 44 = 0$ के अनुप्रस्थ तथा संयुग्मी अक्ष क्रमशः T तथा C है। तो परवलय $x^2 = y + 4$ के Åij]vuqLFkkv {kT के नीचे तथा संयुग्मी अक्ष की दांयी तरफ के क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

   (A) $4\sqrt{6} + \frac{44}{3}$ (B) $4\sqrt{6} + \frac{28}{3}$ (C) $4\sqrt{6} - \frac{44}{3}$ (D) $4\sqrt{6} - \frac{28}{3}$

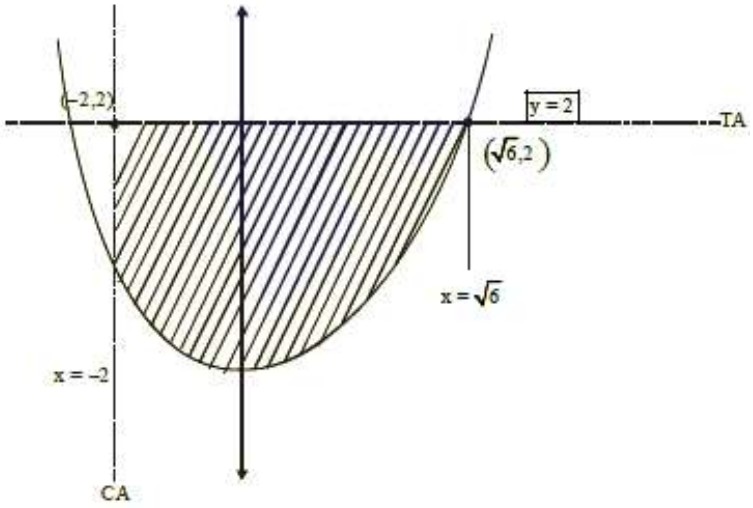
Ans. B

Sol. $16(x^2 + 4x) - (y^2 - 4y) + 44 = 0$

$16(x + 2)^2 - 64 - (y - 2)^2 + 4 + 44 = 0$

$16(x + 2)^2 - (y - 2)^2 = 16$

$$\frac{(x+2)^2}{1}-\frac{(y-2)^2}{16}=1$$



$$A=\int_{-2}^{\sqrt{6}}\left(2-\left(x^2-4\right)\right)dx$$

$$A=\int_{-2}^{\sqrt{6}}\left(6-x^2\right)dx=\left(6x-\frac{x^3}{3}\right)_{-2}^{\sqrt{6}}$$

$$A=\left(6\sqrt{6}-\frac{6\sqrt{6}}{3}\right)-\left(-12+\frac{8}{3}\right)$$

$$A=\frac{12\sqrt{6}}{3}+\frac{28}{3}$$

$$A=4\sqrt{6}+\frac{28}{3}$$

3. Let $A=\left\{(x,y)\in IR^2 : y\geq 0, 2x\leq y\leq \sqrt{4-(x-1)^2}\right.$ and

$B=\left\{(x,y)\in IR\times IR : 0\leq y\leq \min\left\{2x,\sqrt{4-(x-1)^2}\right\}\right\}$.

Then the ratio of the area of A to the area of B is : **[JEE(Main)Jan.2023]**

माना $A=\left\{(x,y)\in IR^2 : y\geq 0, 2x\leq y\leq \sqrt{4-(x-1)^2}\right.$ और

$B=\left\{(x,y)\in IR\times IR : 0\leq y\leq \min\left\{2x,\sqrt{4-(x-1)^2}\right\}\right\}$ है।

तब A के क्षेत्रफल तथा B के क्षेत्रफल का अनुपात है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi+1}{\pi-1}$ (B) $\frac{\pi}{\pi+1}$ (C) $\frac{\pi}{\pi-1}$ (D) $\frac{\pi-1}{\pi+1}$

Ans. D

Sol. $2x \le y \le \sqrt{4-(x-1)^2} : y \ge 0$

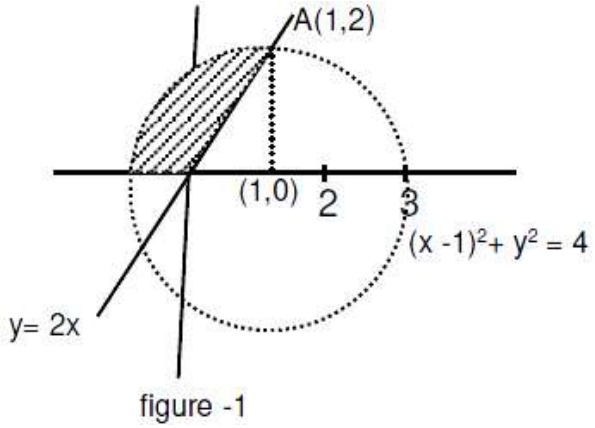


figure -1

area of A : area of shaded Region as show in figure–1

$\therefore$ area of $A = \dfrac{\pi \times 4}{2} - \left(\dfrac{1}{2} \times 1 \times 2\right)$

$= \pi - 1$

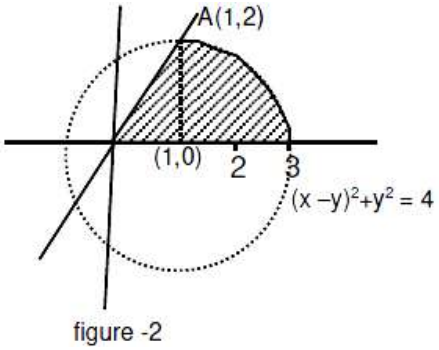


figure -2

area of B : area of shaded Region in as show in above figure–2

$\therefore$ area of $B = \dfrac{1}{2} \times 1 \times 2 + \dfrac{\pi \times 4}{4} = (1+\pi)$

$\dfrac{A}{B} = \dfrac{\text{area of A}}{\text{area of B}} = \dfrac{\pi - 1}{\pi + 1}$

4. Let Δ be the area of the region $\left\{(x,y) \in IR^2 : x^2 + y^2 \le 21, y^2 \le 4x, x \ge 1\right\}$.

Then $\dfrac{1}{2}\left(\Delta - 21\sin^{-1}\dfrac{2}{\sqrt{7}}\right)$ is equal to : **[JEE(Main)Jan.2023]**

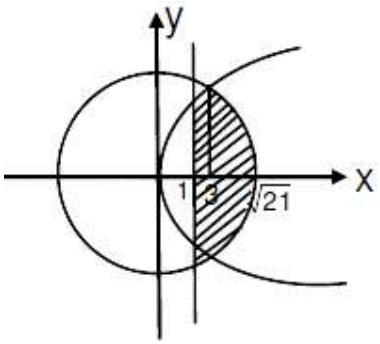
माना क्षेत्र $\left\{(x,y) \in IR^2 : x^2 + y^2 \le 21, y^2 \le 4x, x \ge 1\right\}$ का क्षेत्रफल Δ है, तब $\dfrac{1}{2}\left(\Delta - 21\sin^{-1}\dfrac{2}{\sqrt{7}}\right)$ बराबर है :

[JEE(Main)Jan.2023]

(A) $2\sqrt{3}-\frac{2}{3}$ (B) $\sqrt{3}-\frac{2}{3}$ (C) $2\sqrt{3}-\frac{1}{3}$ (D) $\sqrt{3}-\frac{4}{3}$

Ans. D

Sol.



$$\text{Required area} = 2\left(\int_1^2 2\sqrt{x}dx + \int_3^{\sqrt{21}}\sqrt{21-x^2}dx\right) = 2\left(2\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^3 + \left[\frac{x}{2}\sqrt{21-x^2}+\frac{21}{2}\sin^{-1}\left(\frac{x}{\sqrt{21}}\right)\right]_3^{\sqrt{21}}\right)$$

$$2\left(\frac{4}{3}\left(3\sqrt{3}-1\right)+\frac{21\pi}{4}-\frac{3}{2}\times 2\sqrt{3}-\frac{21}{2}\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)\right) = 8\sqrt{3}-\frac{8}{3}+\frac{21\pi}{2}-6\sqrt{3}-21\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$$

$$\therefore \Delta = 2\sqrt{3}+\frac{21\pi}{2}-\frac{8}{3}-21\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$$

$$\therefore \frac{1}{2}\left(\Delta - 21\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\right)$$

$$\frac{1}{2}\left[2\sqrt{3}-\frac{8}{3}+\frac{21\pi}{2}-21\left(\sin^{-1}\left(\frac{\sqrt{3}}{7}\right)+\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\right)\right] \qquad \because \sin^{-1}\left(\frac{\sqrt{3}}{\sqrt{7}}\right)+\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)=\frac{\pi}{2}$$

$$\frac{1}{2}\left(\Delta - 21\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\right) = \frac{1}{2}\left[2\sqrt{3}-\frac{8}{3}+\frac{21\pi}{2}-21\left(\frac{\pi}{2}\right)\right] = \sqrt{3}-\frac{4}{3}$$

5. The area of the region $A=\left\{(x,y):|\cos x-\sin x|\le y\le \sin x, 0\le x\le \frac{\pi}{2}\right\}$ is : **[JEE(Main)Jan.2023]**

क्षेत्र $A=\left\{(x,y):|\cos x-\sin x|\le y\le \sin x, 0\le x\le \frac{\pi}{2}\right\}$ का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $\sqrt{5}-2\sqrt{2}+1$ (B) $1-\frac{3}{\sqrt{2}}+\frac{4}{\sqrt{5}}$

(C) $\frac{3}{\sqrt{5}}-\frac{3}{\sqrt{2}}+1$ (D) $\sqrt{5}+2\sqrt{2}-4.5$

Ans. A

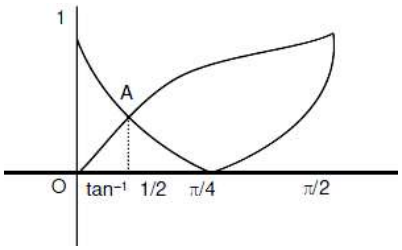
Sol. $$y=\begin{cases}\cos x-\sin x & : \quad 0\le x\le\frac{\pi}{4}\\ \sin x-\cos x & : \quad \frac{\pi}{4}<x\le\frac{\pi}{2}\end{cases}$$

$$y^1=\begin{cases}-\sin x-\cos x & : \quad 0\le x\le\frac{\pi}{4}\\ \cos x+\sin x & : \quad \frac{\pi}{4}<x\le\frac{\pi}{2}\end{cases}$$

For pt. A

$\cos x-\sin x=\sin x$

$\tan x=\frac{1}{2}$



Let $x=\tan^{-1}\frac{1}{2}$

Area $=\int_{\tan^{-1}1/2}^{\pi/4}(\sin x-(\cos x-\sin x))dx+\int_{\pi/4}^{\pi/2}(\sin x-\sin x+\cos x)dx$

$=\left(-2\cos x-\sin x\right)_x^{\pi/4}+\left(\sin x\right)_{\pi/4}^{\pi/2}$

$=\left(\frac{-2}{\sqrt{2}}-\frac{1}{\sqrt{2}}\right)-\left(-2\cos x-\sin x\right)+\left(1-\frac{1}{\sqrt{2}}\right)$

$=-\sqrt{2}-\frac{1}{\sqrt{2}}+2\cos x+\sin x+1-\frac{1}{\sqrt{2}}=-2\sqrt{2}+1+\frac{2\times2}{\sqrt{5}}+\frac{1}{\sqrt{5}}$

$=\sqrt{5}+1-2\sqrt{2}$

6. Let q be the maximum integral value of p in [0,10] for which the roots of the equation $x^2-px+\frac{5}{4}p=0$ are rational. Then the area of the region $\left\{(x,y):0\le y\le(x-q)^2,0\le x\le q\right\}$ is : **[JEE(Main)Jan.2023]**

माना [0,10] में p का अधिकतम पूर्णांक मान, जिसके लिए समीकरण $x^2 - px + \frac{5}{4}p = 0$ के मूल परिमेय q है। तब क्षेत्र $\{(x,y) : 0 \le y \le (x-q)^2, 0 \le x \le q\}$ का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) 25 (B) $\frac{125}{3}$ (C) 164 (D) 243

Ans. D

Sol. $D = p^2 - 5p$ = Perfect square of a rational number

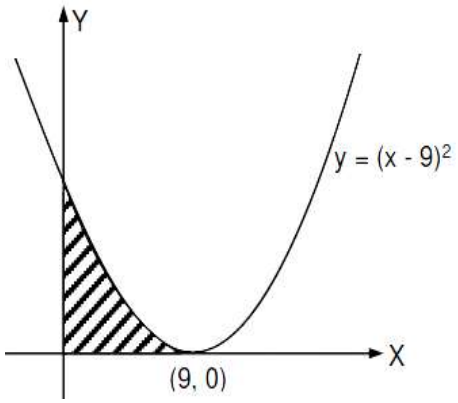
$\Rightarrow D = p(p-5)$

Noe put values in reverse orde as q is maximum value of p

p = 10 reject

p = 9 select

$\Rightarrow q = 9$



So area of shaded region $= \int_0^9 (x-9)^2 dx$

$= \left[\frac{(x-9)^3}{3}\right]_0^9$

$\Rightarrow$ 243 sq. unit

7. The area enclosed by the closed curve C given by the differential equation $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$ is 4π. Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is : **[JEE(Main)Jan.2023]**

अवकल समीकरण $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$ से प्राप्त संवृत्त वक्र C से घिरे क्षेत्र का क्षेत्रफल 4π है। माना वक्र C व y अक्ष के प्रतिच्छेदन बिंदु P व Q है। यदि वक्र C के बिंदु P व Q पर अभिलम्ब x अक्ष को क्रमशः बिंदु R व S पर मिलते है, तब रेखा खण्ड RS की y Ecbz S **[JEE(Main)Jan.2023]**

(A) $\frac{2\sqrt{3}}{3}$ (B) 2 (C) $2\sqrt{3}$ (D) $\frac{4\sqrt{3}}{3}$

Ans. D

Sol. $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$

$\frac{dy}{dx} = \frac{x+a}{2-y}$

$(2-y)\,dy = (x+a)\,dx$

$2y - \frac{y^2}{2} = \frac{x^2}{2} + ax + c$

$a+c=-\frac{1}{2}$ as $y(1)=0$

$X^2+Y^2+2ax-4y-1-2a=0$

$\pi r^2=4\pi$

$r^2=4$

$4=a^2+4+1+2a$

$(a+1)^2=0$

P or $Q=\left(0,2\pm\sqrt{3}\right)$

Equation of normal at P, Q are $y-2=\sqrt{3}(x-1)$

$y-2=-\sqrt{3}(x-1)$

$R=\left(1-\frac{2}{\sqrt{3}},0\right)$

$S=\left(1+\frac{2}{\sqrt{3}},0\right)$

$RS=\frac{4}{\sqrt{3}}=4\frac{\sqrt{3}}{3}$

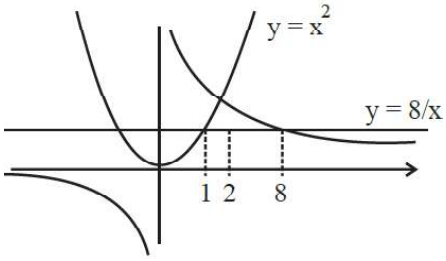
8. The area of the region given by $\left\{(x,y):xy\le 8,1\le y\le x^2\right\}$ is : **[JEE(Main)Jan.2023]**

$\left\{(x,y):xy\le 8,1\le y\le x^2\right\}$ द्वारा दिए क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $8\log_e 2+\frac{7}{6}$ (B) $16\log_e 2-\frac{14}{3}$ (C) $8\log_e 2-\frac{13}{3}$ (D) $16\log_e 2+\frac{7}{3}$

Ans. B

Sol. $xy\le 8$ $1\le y\le x^2$



Area $=\int_1^2\left(x^2-1\right)dx+\int_2^8\left(\frac{8}{x}-1\right)dx$

$=\left(\frac{x^3}{3}\right)_1^2+8\left(\ell nx\right)_2^8-\left(x\right)_1^8$

$$=\frac{7}{3}+8\left(2\ell n2\right)-7$$

$$=16\ell n2-\frac{14}{3}$$

9. If the area of the region bounded by the curves $y^2 - 2y = -x$, $x + y = 0$ is A, then 8 A is equal to _____.

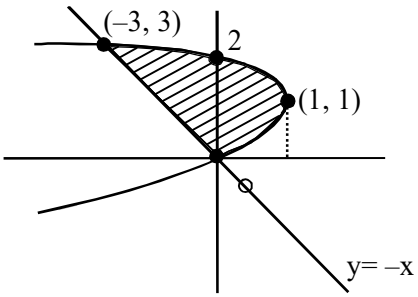
[JEE(Main)Jan.2023]

यदि वक्रों $y^2 - 2y = -x$, $x + y = 0$ से घिरे क्षेत्र का क्षेत्रफल A है, तो 8 A बराबर है _____

[JEE(Main)Jan.2023]

Ans. 36

Sol.



$$A=\int_0^3\left[\left(2y-y^2\right)-(-y)\right]dy=\int_0^3\left(3y-y^2\right)dy=\left[\frac{3y^2}{2}-\frac{y^3}{3}\right]_0^3$$

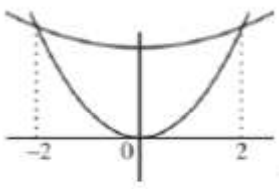
$$=\frac{3\times9}{2}-\frac{27}{3}=\frac{9}{2}$$ So, 8A = 36

10. If the area enclosed by the parabolas $P_1 : 2y = 5x^2$ and $P_2 : x^2 - y + 6 = 0$ is equal to the area enclosed by P_1 and $y = \alpha x$, $\alpha > 0$, then α^3 is equal to _____. **[JEE(Main)Jan.2023]**

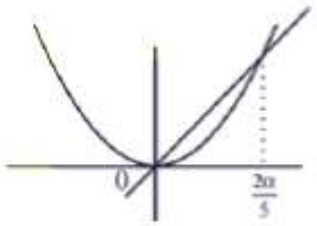
यदि परवलयों $P_1 : 2y = 5x^2$ तथा $P_2 : x^2 - y + 6 = 0$ से घिरे क्षेत्र का क्षेत्रफल, परवलय P_1 तथा $y = \alpha x$, $\alpha > 0$ से घिरे क्षेत्र के {kslkQy d scjkcj g\$r ksα^3 है_____ **[JEE(Main)Jan.2023]**

Ans. 600

Sol.



Abscissa of point of intersection of $2y = 5x^2$
and $y = x^2 + 6$ is ± 2



$$\text{Area } = 2\int_0^2\left(x^2+6-\frac{5x^2}{2}\right)dx = \int_0^{\frac{2\alpha}{5}}\left(\alpha x-\frac{5x^2}{2}\right)dx$$

$$\Rightarrow \int_0^{\frac{2\alpha}{5}}\left(\alpha x-\frac{5x^2}{2}\right)dx = 16$$

$$\Rightarrow \alpha^3 = 600$$

11. Let α be the area of the larger region bounded by the curve $y^2 = 8x$ and the lines $y = x$ and $x = 2$, which lies in the first quadrant. Then the value of 3α is equal to _____. **[JEE(Main)Jan.2023]**

माना प्रथम चतुर्थांश में वक्र $y^2 = 8x$ और रेखाओं $y = x$ एवं $x = 2$ से घिरे बड़े क्षेत्र का क्षेत्रफल α है। तो 3α का मान बराबर है _____ **[JEE(Main)Jan.2023]**

Ans. 22

Sol. On solving $y^2 = 8x$ and $y = x$, we get

$x = 0, 8$

Now shaded area $= \int_2^8\left(2\sqrt{2}\cdot\sqrt{x}-x\right)dx$

$$= 2\sqrt{2}\int_2^8\sqrt{x}\cdot dx-\int_2^8 x\,dx$$

$$= \frac{4\sqrt{2}}{3}\left[x^{\frac{3}{2}}\right]_2^8-\frac{1}{2}\left[x^2\right]_2^8$$

$$= \frac{4\sqrt{2}}{3}\left[8^{\frac{3}{2}}-2^{\frac{3}{2}}\right]-\frac{1}{2}\left[8^2-2\right]$$

$$= \frac{4\sqrt{2}\times 2\sqrt{2}}{3}[8-1]-\frac{1}{2}\times 60$$

$$= \frac{8\times 2\times 7}{3}-30 = \frac{22}{3} = \alpha$$

$$3\alpha = 22$$

12. Let A be the area of the region $\left\{(x,y): y\ge x^2, y\ge(1-x)^2, y\le 2x(1-x)\right\}$. Then 540 A is equal to _____. **[JEE(Main)Jan.2023]**

माना क्षेत्र $\left\{(x,y): y \geq x^2, y \geq (1-x)^2, y \leq 2x(1-x)\right\}$ का क्षेत्रफल A है। तब 540 A का मान बराबर है_____

[JEE(Main)Jan.2023]

Ans. 25

Sol. $A = 2\int_{\frac{1}{3}}^{\frac{1}{2}} \left(2x - 2x^2 - (1-x)^2\right)dx$

$= 2\left[2x^2 - x^3 - x\right]_{1/3}^{1/2}$

So $A = \frac{5}{108} \Rightarrow 540A = \frac{5}{108} \times 540 = 25$

13. Let for $x \in R$ $f(x) = \frac{x+|x|}{2}$ and $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$. The area bounded by the curve $y = (fog)(x)$ and the lines $y = 0$, $2y - x = 15$ is equal to _____.

[JEE(Main)Jan.2023]

माना $x \in R$ के लिए $f(x) = \frac{x+|x|}{2}$ तथा $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ है, तो वक्र $y = (fog)(x)$ तथा रेखाओं $y = 0$, $2y - x = 15$ से घिरे क्षेत्र का क्षेत्रफल बराबर है_____

[JEE(Main)Jan.2023]

Ans. 72

Sol. $f(x) = \frac{x+|x|}{2} = \begin{bmatrix} x & x \geq 0 \\ 0 & x < 0 \end{bmatrix}$

$g(x) = \begin{bmatrix} x^2 & x \geq 0 \\ x & x < 0 \end{bmatrix}$

$fog(x) = f[g(x)] = \begin{bmatrix} g(x) & g(x) \geq 0 \\ 0 & g(x) < 0 \end{bmatrix}$

$fog(x) = \begin{bmatrix} x^2 & x \geq 0 \\ 0 & x < 0 \end{bmatrix}$

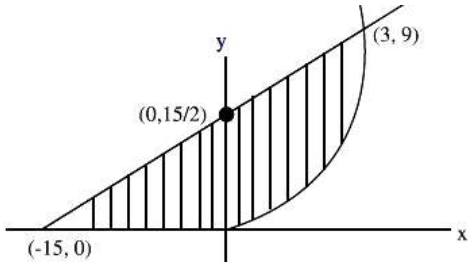
$2y - x = 15$

$A = \int_0^3 \left(\frac{x+15}{2} - x^2\right)dx + \frac{1}{2} \times \frac{15}{2} \times 15$

$\frac{x^2}{4} + \frac{15x}{2} - \frac{x^3}{3}\Big|_0^3 + \frac{225}{4}$

$$= \frac{9}{4} + \frac{45}{2} - 9 + \frac{225}{4} = \frac{99 - 36 + 225}{4}$$

$$= \frac{288}{4} = 72$$



14. Let the area of the region $\left\{(x, y) : |2x - 1| \le y \le \left|x^2 - x\right|, 0 \le x \le 1\right\}$ be A. Then $(6A + 11)^2$ is equal to _____ .

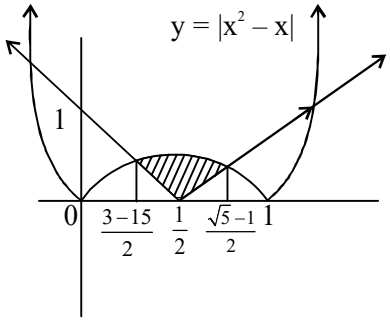
[JEE(Main)Jan.2023]

माना क्षेत्र $\left\{(x, y) : |2x - 1| \le y \le \left|x^2 - x\right|, 0 \le x \le 1\right\}$ का क्षेत्रफल A है। तो $(6A + 11)^2$ बराबर है_____

[JEE(Main)Jan.2023]

Ans. 125

Sol. $y \ge |2x - 1|$ $y \le |x^2 - x|$



Both curve are symmetric about $x = \frac{1}{2}$ Hence

$$A = 2\int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} \left(\left(x - x^2\right) - \left(1 - 2x\right)\right) dx$$

$$A = 2\int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} \left(-x^2 + 3x - 1\right) dx = 2\left(\frac{-x^3}{3} + \frac{3}{2}x^2 - x\right)_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}}$$

on solving $6A + = 5\sqrt{5}$

$(6A + 11)^2 = 125$

15. Let A be the area bounded by the curve $y = x|x-3|$, the x-axis and the ordinates $x = -1$ and $x = 2$. Then 12A is equal to _____. **[JEE(Main)Jan.2023]**

माना वक्र $y = x|x-3|$, x-अक्ष तथा कोटियों $x = -1$ व $x = 2$ से घिरे क्षेत्र का क्षेत्रफल A है। तब 12A का मान है_____ **[JEE(Main)Jan.2023]**

Ans. 62

Sol. $A = \int_{-1}^{0} \left(x^2 - 3x\right)dx + \int_{0}^{2} \left(3x - x^2\right)dx$

$$\Rightarrow A = \frac{x^3}{3} - \frac{3x^2}{2}\Big|_{-1}^{0} + \frac{3x^2}{2} - \frac{x^3}{3}\Big|_{0}^{2}$$

$$\Rightarrow A = \frac{11}{6} + \frac{10}{3} = \frac{31}{6}$$

$\therefore$ $12A = 62$

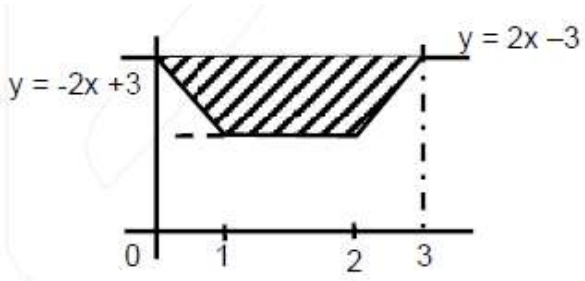
16. The area bounded by the curves $y = |x-1| + |x-2|$ and $y = 3$ is equal to : **[JEE(Main)April,2023]**

वक्रों $y = |x-1| + |x-2|$ तथा $y = 3$ से परिबद्ध क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) 4 (B) 3 (C) 6 (D) 5

Ans. A

Sol.



$$\text{area} = \frac{1}{2} \times (1+3) \times 2 = 4$$

17. The area of the region $\left\{(x, y) : x^2 \le y \le 8 - x^2, y \le 7\right\}$ is : **[JEE(Main)April,2023]**

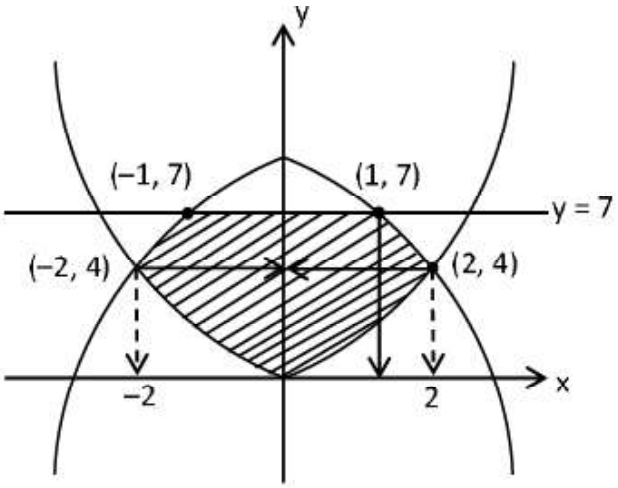
क्षेत्र $\left\{(x, y) : x^2 \le y \le 8 - x^2, y \le 7\right\}$ का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) 18 (B) 24 (C) 20 (D) 21

Ans. C

Sol. $y \ge x^2 \quad y \le 8 - x^2 \quad y \le 7$

$x^2 = 8 - x^2$



$x^2 = 4$

$x = \pm 2$

$2\left(1.7 + \int_1^2 \left(8 - 2x^2\right)dx\right) - 2\int_0^1 \left(x^2\right)dx$

$= 2\left[7 + \left(8x - \frac{2x^3}{3}\right)_1^2\right] - 2\left(\frac{x^3}{3}\right)_0^1$

$= 2\left[7 + \left(16 - \frac{16}{3}\right) - \left(8 - \frac{2}{3}\right)\right] - 2\left(\frac{1}{3}\right)$

$= 2\left[7 + \frac{32}{3} - \frac{22}{3}\right] = 2\left[7 + \frac{10}{3}\right]\frac{-2}{3}$

$= \frac{60}{3} = 20$

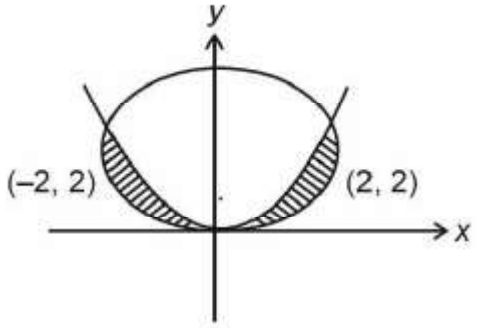
18. Area of the region $\left\{(x, y) : x^2 + (y-2)^2 \le 4, x^2 \ge 2y\right\}$ is : **[JEE(Main)April,2023]**

क्षेत्र $\left\{(x, y) : x^2 + (y-2)^2 \le 4, x^2 \ge 2y\right\}$ का क्षेत्रफल : **[JEE(Main)April,2023]**

(A) $2\pi - \frac{16}{3}$ (B) $\pi + \frac{8}{3}$ (C) $\pi - \frac{8}{3}$ (D) $2\pi + \frac{16}{3}$

Ans. A

Sol. $x^2 + (y-2)^2 \le 4, x^2 \ge 2y$



Area of required region $= 2\left[\frac{1}{4}\pi(4) - \int_0^2 \sqrt{2}\cdot\sqrt{y}dy\right]$

$\Rightarrow 2\left[\pi - \frac{\sqrt{2}\cdot y^{3/2}}{3/2}\Big|_0^2\right]$

$\Rightarrow 2\left[\pi - \frac{2\sqrt{2}}{3}\cdot 2\sqrt{2}\right] = 2\left[\pi - \frac{8}{3}\right] = 2\pi - \frac{16}{3}$

19. The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is :

[JEE(Main)April,2023]

वक्र $y = x^3$ तथा इसके बिंदु $(-1, -1)$ पर स्पर्श रेखा से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2023]**

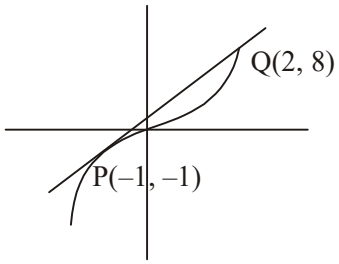
(A) $\frac{23}{4}$ (B) $\frac{19}{4}$ (C) $\frac{27}{4}$ (D) $\frac{31}{4}$

Ans. C

Sol. Given $y = x^3$ _____(1)

$\Rightarrow \frac{dy}{dx} = 3x^2$

$\left(\frac{dy}{dx}\right)_{(-1,-1)} = 3$



Equation of tangent at $(-1, -1)$

$(y + 1) = 3(x + 1)$

$y = 3x + 2$ _____(2)

Solving (1) and (2)

$x^3 = 3x + 2$

$x^3 - 3x - 2 = 0$: -1, -1, α

$-2 + \alpha = 0$

$\alpha = 2$

$Q \equiv (2, 8)$

Required area $= \int_{-1}^{2} \left(3x + 2 - x^3\right) dx$

$= \frac{3}{2}(4-1) + 2(2+1) - \frac{1}{4}(16-1)$

$= \frac{9}{2} + 6 - \frac{15}{4} = \frac{18+24-15}{4} = \frac{27}{4}$

20. The area of the region enclosed by the curve $f(x) = \max\{\sin x, \cos x\}, -\pi \le x \le \pi$ and the x-axis is :

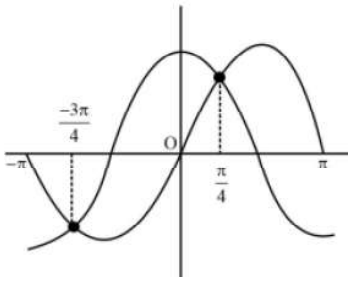
[JEE(Main)April,2023]

वक्र $f(x) = \max\{\sin x, \cos x\}, -\pi \le x \le \pi$ तथा x-अक्ष से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) 4 (B) $2(\sqrt{2}+1)$ (C) $2\sqrt{2}(\sqrt{2}+1)$ (D) $4(\sqrt{2})$

Ans. A

Sol.



Area =

$$\left|\int_{-\pi}^{\frac{-3\pi}{4}} \sin x\,dx\right| + \left|\int_{\frac{-3\pi}{4}}^{\frac{-\pi}{2}} \cos x\,dx\right| + \int_{\frac{-\pi}{2}}^{\frac{\pi}{4}} \cos x\,dx + \int_{\frac{\pi}{4}}^{\pi} \sin x\,dx = 4$$

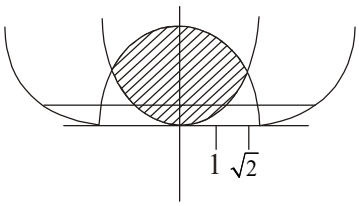
21. The area of the region $\left\{(x,y): x^2 \le y \le |x^2-4|, y \ge 1\right\}$ is : **[JEE(Main)April,2023]**

क्षेत्र $\left\{(x,y): x^2 \le y \le |x^2-4|, y \ge 1\right\}$ का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) $\frac{3}{4}(4\sqrt{2}-1)$ (B) $\frac{4}{3}(4\sqrt{2}+1)$ (C) $\frac{3}{4}(4\sqrt{2}+1)$ (D) $\frac{4}{3}(4\sqrt{2}-1)$

Ans. D

Sol.



$$A = 2\left(\int_0^1 \left(3-x^2\right)dx + \int_1^{\sqrt{2}} \left(4-2x^2\right)dx\right) = 2\left(\left(3x-\frac{x^3}{3}\right)_0^1 + \left(4x-\frac{2x^3}{3}\right)_1^{\sqrt{2}}\right)$$

$$= 2\left(3-\frac{1}{3}+\left(4\sqrt{2}-\frac{4\sqrt{2}}{3}-4+\frac{2}{3}\right)\right) = 2\left(\frac{-2}{3}+\frac{2}{3}\left(4\sqrt{2}\right)\right) = \frac{4\left(4\sqrt{2}-1\right)}{3}$$

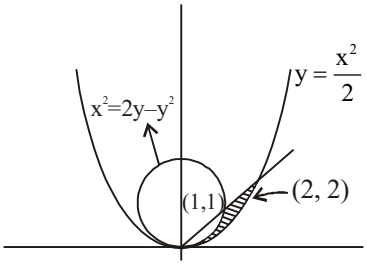
22. If the area of the region $S = \left\{(x,y): 2y-y^2 \le x^2 \le 2y, x \ge y\right\}$ is equal to $\frac{n+2}{n+1} - \frac{\pi}{n-1}$, then the natural number n is equal to _____. **[JEE(Main)April,2023]**

यदि क्षेत्र $S = \left\{(x,y): 2y-y^2 \le x^2 \le 2y, x \ge y\right\}$ का क्षेत्रफल $\frac{n+2}{n+1} - \frac{\pi}{n-1}$ के बराबर है, तब प्राकृतिक संख्या n बराबर है

_____ **[JEE(Main)April,2023]**

Ans. 5

Sol. $2y - y^2 \le x^2 \le 2y, x \ge y$



$$\text{Area} = \int_1^2 (\sqrt{2y} - y)dy + \int_0^1 \left(\sqrt{2y} - \sqrt{2y - y^2}\right)dy$$

$$= \frac{7 - 2^{\frac{5}{2}}}{6} + \frac{2^{\frac{7}{2}} - 3\pi}{12}$$

$$= \frac{14 - 2^{\frac{7}{2}} + 2^{\frac{7}{2}} - 3\pi}{12} = \frac{7}{6} - \frac{\pi}{4}$$

$n = 5$

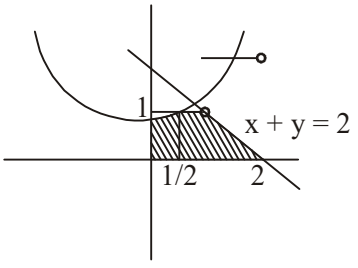
23. Let the area enclosed by the lines $x + y = 2$, $y = 0$, $x = 0$ and the curve $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ where $[x]$ denotes the greatest integer $\le x$, be A. Then the value of 12A is _____. **[JEE(Main)April,2023]**

माना $[x]$ महत्तम पूर्णांक $\le x$ है। माना रेखाओं $x + y = 2$, $y = 0$, $x = 0$ तथा वक्र $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ से घिरे क्षेत्र का {ksk Qy A है। तो 12 A का मान है : **[JEE(Main)April,2023]**

Ans. 17

Sol. $$A = \int_0^{\frac{1}{2}} \left(x^2 + \frac{3}{4}\right)dx + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{2}\right) \times 2$$

$$\frac{1}{3}.\frac{1}{8} + \frac{3}{4}.\frac{1}{2} + 1$$



$$12A = \frac{1}{2} + \frac{9}{2} + 12 = 17$$

24. Let y = p(x) be the parabola passing through the points (–1, 0), (0, 1) and (1, 0). If the area of the region $\left\{(x, y) : (x+1)^2 + (y-1)^2 \le 1, y \le p(x)\right\}$ is A, then $12(\pi - 4A)$ is equal to _____.

[JEE(Main)April,2023]

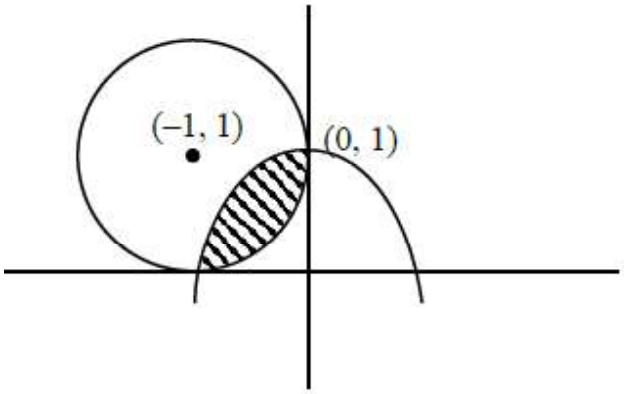
माना बिंदुओं (–1, 0), (0, 1) तथा (1, 0) से होकर जाने वाला परवलय y = p(x) है। यदि क्षेत्र $\left\{(x, y) : (x+1)^2 + (y-1)^2 \le 1, y \le p(x)\right\}$ का क्षेत्रफल A है, तो $12(\pi - 4A)$ बराबर है _____

[JEE(Main)April,2023]

Ans. 16

Sol. There can be infinitely many parabolas through given points.

Let parabola $x^2 = -4a(y-1)$



Passes through (1, 0)

$$\therefore b = -4a(-1) \Rightarrow a = \frac{1}{4}$$

$$\therefore x^2 = -(y-1)$$

Now area covered by parabola $= \int_{-1}^{0} \left(1 - x^2\right) dx$

$$= \left(x - \frac{x^3}{3}\right)_1^b = (0-0) - \left\{-1 + \frac{1}{3}\right\}$$

$$= \frac{2}{3}$$

Required Area = Area of sector – {Area of square – Area covered by Parabola}

$$= \frac{\pi}{4} - \left\{1 - \frac{2}{3}\right\}$$

$$= \frac{\pi}{4} - \frac{1}{3}$$

$$\therefore 12(\pi-4A)=12\left[\pi-4\left(\frac{\pi}{4}-\frac{1}{3}\right)\right]$$

$$=12\left[\pi-\pi+\frac{4}{3}\right]$$

$$=16$$

25. If the area of the region $\left\{(x,y):\left|x^2-2\right|\le y\le x\right\}$ is A, then $6A+16\sqrt{2}$ is equal to _____.

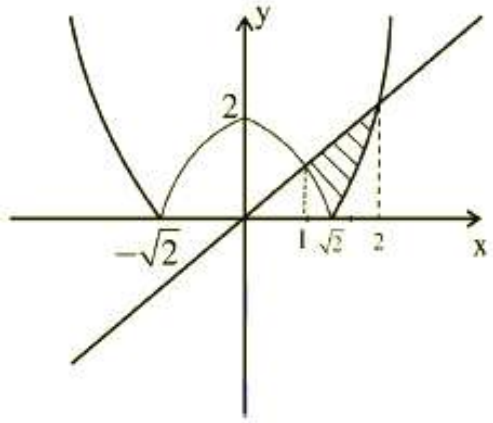
[JEE(Main)April,2023]

यदि क्षेत्र $\left\{(x,y):\left|x^2-2\right|\le y\le x\right\}$ का क्षेत्रफल A है, तो $6A+16\sqrt{2}$ बराबर है _____

[JEE(Main)April,2023]

Ans. 27

Sol. $\left|x^2-2\right|\le y\le x$



$$A=\int_1^{\sqrt{2}}\left(x-\left(2-x^2\right)\right)dx+\int_{\sqrt{2}}^{2}\left(x-\left(x^2-2\right)\right)dx$$

$$=\left(1-2\sqrt{2}+\frac{2\sqrt{2}}{3}\right)-\left(\frac{1}{2}-2+\frac{1}{3}\right)+\left(2-\frac{8}{3}+4\right)-\left(1-\frac{2\sqrt{2}}{3}+2\sqrt{2}\right)$$

$$=-4\sqrt{2}+\frac{4\sqrt{2}}{3}+\frac{7}{6}+\frac{10}{3}=\frac{-8\sqrt{2}}{3}+\frac{9}{2}$$

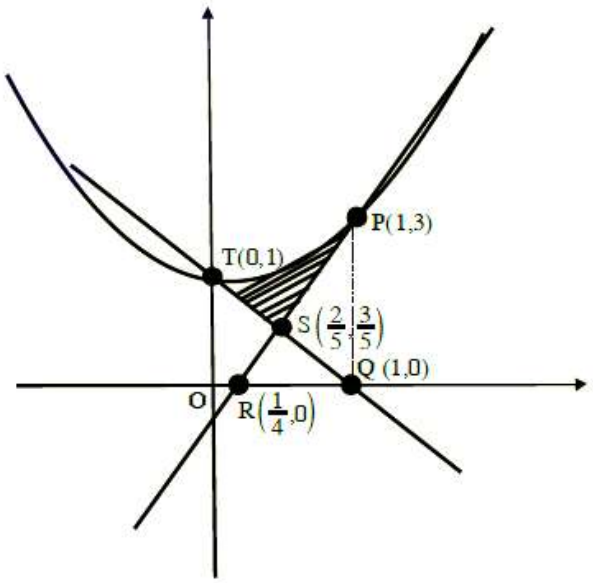
$$6A=-16\sqrt{2}+27\ \therefore 6A+16\sqrt{2}=27$$

26. If A is the area in the first quadrant enclosed by the curve $C: 2x^2-y+1=0$, the tangent to C at the point (1, 3) and the line $x+y=1$, then the value of 60A is _____. **[JEE(Main)April,2023]**

यदि वक्र $C: 2x^2-y+1=0$, C के बिंदु (1, 3) पर स्पर्श रेखा तथा रेखा $x+y=1$ से घिरे, प्रथम चतुर्थांश में, क्षेत्र का क्षेत्रफल A है, तो 60A बराबर है _____ **[JEE(Main)April,2023]**

Ans. 16

Sol.



$y = 2x^2 + 1$

Tangent at (1, 3)

$y = 4x - 1$

$$A = \int_0^1 \left(2x^2 + 1\right)dx - \text{area of}(\Delta QOT) - \text{are of}(\Delta PQR) + \text{area of}(\Delta QRS)$$

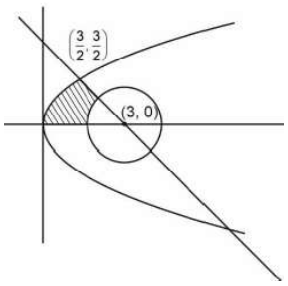
$$A = \left(\frac{2}{3} + 1\right) - \frac{1}{2} - \frac{9}{8} + \frac{9}{40} = \frac{16}{60}$$

27. If the area bounded by the curve $2y^2 = 3x$, lines $x + y = 3$, $y = 0$ and outside the circle $(x - 3)^2 + y^2 = 2$ is A, then $4(\pi + 4A)$ is equal to _____. **[JEE(Main)April,2023]**

यदि वक्र $2y^2 = 3x$, रेखाओं $x + y = 3$, $y = 0$ तथा वृत्त $(x - 3)^2 + y^2 = 2$ के बाहर गिरा हुआ क्षेत्रफल A है, तो $4(\pi + 4A)$ मान gkxk_____ **[JEE(Main)April,2023]**

Ans. 42

Sol.



$$\int_0^{3/2}\left((3-y) - \frac{2y^2}{3}\right)dy - \pi\left(\sqrt{2}\right)^2 \cdot \frac{1}{8}$$

$$= \frac{36 - 9 - 6}{8} - \frac{\pi}{4} = \frac{21}{8} - \frac{\pi}{4}$$

$4(\pi + 4A) = 42$

28. The area (in square units) of the region bounded by the parabola $y^2 = 4(x-2)$ and the line $y = 2x - 8$, is :

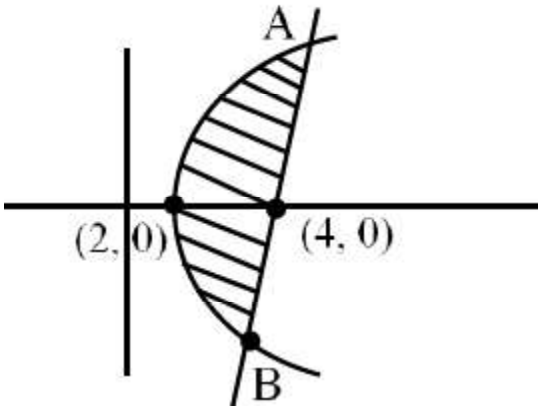
[JEE(Main)Jan.2024]

परवलय $y^2 = 4(x-2)$ तथा रेखा $y = 2x - 8$ से घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाई में) है : **[JEE(Main)Jan.2024]**

(A) 9 (B) 8 (C) 7 (D) 6

Ans. A

Sol.



$y^2 = 4x - 8$

$y = 2x - 8$

$\Rightarrow y^2 = 2(y+8) - 8$

$\Rightarrow y^2 - 2y - 8 = 0$

$y = 4 \qquad y = -2$

$$\int_{-2}^{4}\left(\frac{y+8}{2} - \frac{y^2+8}{4}\right)dy$$

$$\Rightarrow \int_{-2}^{4}\frac{y}{2}dy - \int_{-2}^{4}\frac{y^2}{4}dy + \int_{-2}^{4}2dy$$

$$\Rightarrow \left[\frac{y^2}{4}\right]_{-2}^{4} - \frac{1}{4}\left[\frac{y^3}{3}\right]_{-2}^{4} + 2[y]_{-2}^{4} = 9$$

29. The area of the region $\left\{(x,y): y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$ is :

[JEE(Main)Jan.2024]

क्षेत्र $\left\{(x,y): y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$ का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) $\frac{32}{3}$ (B) $\frac{16}{3}$ (C) $\frac{64}{3}$ (D) $\frac{8}{3}$

Ans. A

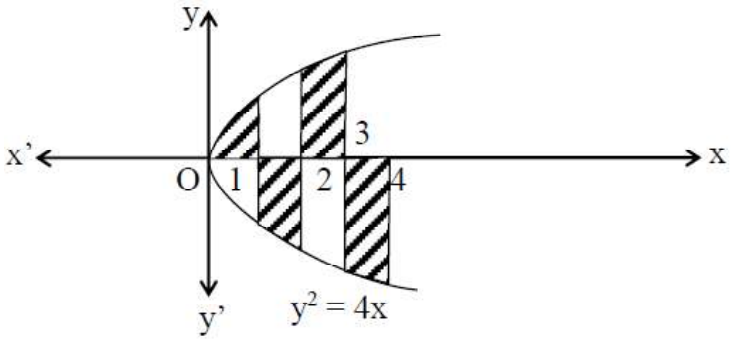
Sol. $\left\{(x,y); y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$

$y^2 \le 4x,\ x < 4,\ \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0$

Case (i) $y > 0$ $\Rightarrow \frac{x(x-1)(x-2)}{(x-3)(x-4)} < 0$

$x \in (1,2) \cup (3,4)$

$$\text{Area} = \int_0^1 2\sqrt{x}\,dx + \left|\int_1^2 2\sqrt{x}\,dx\right| + \int_2^3 2\sqrt{x}\,dx + \left|\int_3^4 2\sqrt{x}\,dx\right|$$



$$= \frac{4}{3}\left(\left[x^{\frac{3}{2}}\right]_0^1 + \left|\left[x^{\frac{3}{2}}\right]_1^2\right| + \left[x^{\frac{3}{2}}\right]_2^3 + \left|\left(x^{\frac{3}{2}}\right)_3^4\right|\right)$$

$$A = \frac{4}{3}\left(1 + 2\sqrt{2} - 1 + 3\sqrt{3} - 2\sqrt{2} + 4\sqrt{4} - 3\sqrt{3}\right) = \frac{4}{3} \times 8 = \frac{32}{3}$$

30. The area of the region enclosed by the parabolas $y = 4x - x^2$ and $3y = (x-4)^2$ is equal to :

[JEE(Main)Jan.2024]

परवलयों $y = 4x - x^2$ तथा $3y = (x-4)^2$ से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) 32/9 (B) 4 (C) 6 (D)14/3

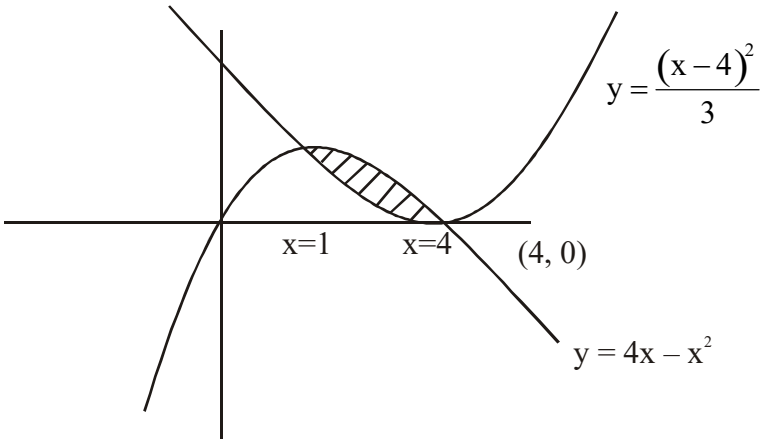
Ans. C

Sol. Solving $3y = (x-4)^2$ and $y = 4x - x^2$

$12x - 3x^2 = x^2 - 8x + 16$

$\Rightarrow 4x^2 - 20x + 16 = 0$

$(x-1)(x-4) = 0 \Rightarrow x = 1, 4$



$$\text{Area} = \int_1^4 \left(\left(4x - x^2\right) - \frac{(x-4)^2}{3} \right) dx$$

$$= \left(2x^2 - \frac{x^3}{3} - \frac{(x-4)^3}{9} \right)_1^4 = 2(16-1) - \frac{1}{3}(3)(21) - \frac{1}{9}(0+27) = 30 - 21 - 3 = 6$$

31. The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to : **[JEE(Main)Jan.2024]**

वक्रों $xy + 4y = 16$ तथा $x + y = 6$ से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) $28 - 30 \log_e 2$ (B) $32 - 30 \log_e 2$ (C) $30 - 28\log_e 2$ (D) $30 - 32 \log_e 2$

Ans. D

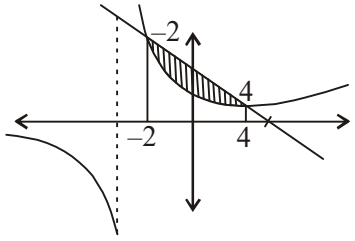
Sol. $y(x + 4) = 16$ & $y = 6 - x$

solve both curves,

$(6 - x)(x + 4) = 16$

$6x - x^2 + 24 - 4x = 16$

$x^2 - 2x - 8 = 0 \Rightarrow x = -2, 4$



$$\int_{-2}^{4} \left(6 - x - \left(\frac{16}{x+4} \right) \right) dx$$

$$\Rightarrow \left[6x - \frac{x^2}{2} - 16\ell n|x+4| \right]_{-2}^{4}$$

$$\Rightarrow (24 - 8 - 16\ell n8) - (-12 - 2 - 16\ell n2)$$

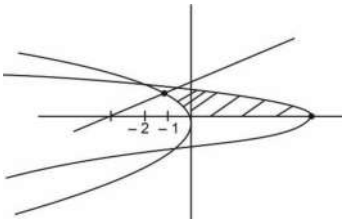
$$\Rightarrow 30 - 32\ell n2$$

32. Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then m + n is equal to _____. **[JEE(Main)Jan.2024]**

माना क्षेत्र $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ का क्षेत्रफल $\frac{m}{n}$ है, जहाँ m तथा n असहभाज्य संख्याएँ हैं। तो m + n बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 119

Sol.



Area between these curve will be

$$\int_{-2}^{-1}\left(-\sqrt{\frac{-x}{2}} + \frac{x+4}{2}\right)dx + \int_{-1}^{0}\left(\sqrt{\frac{8-x}{4}} - \sqrt{\frac{-x}{2}}\right)dx + \int_{0}^{8}\sqrt{\frac{8-x}{4}}dx = \frac{107}{12} = \frac{m}{n}$$

$$\Rightarrow m + n = 119$$

33. If the area of the region $\left\{(x, y) : 0 \leq y \leq \min\left\{2x, 6x - x^2\right\}\right\}$ is A, then 12 A is equal to _____.

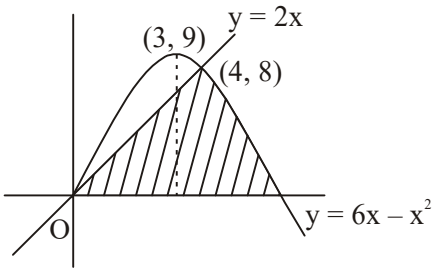
[JEE(Main)Jan.2024]

यदि क्षेत्र $\left\{(x, y) : 0 \leq y \leq \min\left\{2x, 6x - x^2\right\}\right\}$ का क्षेत्रफल A है, तो 12 A बराबर है _____

[JEE(Main)Jan.2024]

Ans. 304

Sol.



Solving $y = 2x$ and $y = 6x - x^2$

$2x = 6x - x^2$

$x^2 - 4x = 0$

$y = x = 0, x = 4, y = 8$

$A = \frac{1}{2}4.8 + \int_4^6 \left(6x - x^2\right)dx$

$= 16 + \left(3x^2 - \frac{x^3}{3}\right)_4^6$

$A = 16 + 3(36 - 16) - \frac{1}{3}(216 - 64)$

$A = 16 + 60 - \frac{152}{3}$

$A = 16 + \frac{28}{3}$

$12A = 304$

34. The area (in sq. units) of the part of the circle $x^2 + y^2 = 169$ which is below the line $5x - y = 13$ is $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$, where α, β are coprime number. Then $\alpha + \beta$ is equal to _____.

[JEE(Main)Jan.2024]

वृत्त $x^2 + y^2 = 169$ के रेखा $5x - y = 13$ से नीचे के भाग का क्षेत्रफल (वर्ग इकाई में) $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$ है, जहां α, β असहभाज्य संख्याऐं है। तो $\alpha + \beta$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 171

Sol. Solving line and circle $x = 0, 5$

$\tan\alpha = \frac{12}{5}$

$\sin\alpha = \frac{12}{13}$

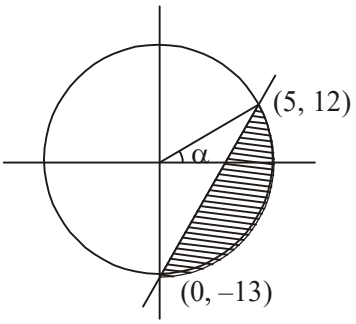
$\sin\left(\frac{\pi}{2} + \alpha\right) = \cos\alpha = \frac{5}{13}$

Now area of segment $= \frac{1}{2}\left(r^2\right)(\theta - \sin\theta)$

$$= \frac{1}{2} \times 169 \times \left[\frac{\pi}{2} + \sin^{-1} \alpha - \sin\left(\frac{\pi}{2} + \alpha \right) \right]$$

$$= \frac{169}{2} \left[\frac{\pi}{2} + \sin^{-1} \frac{12}{13} - \frac{5}{13} \right]$$

$$= \frac{169\pi}{4} + \frac{169}{2} \sin^{-1} \frac{12}{13} - \frac{65}{2}$$



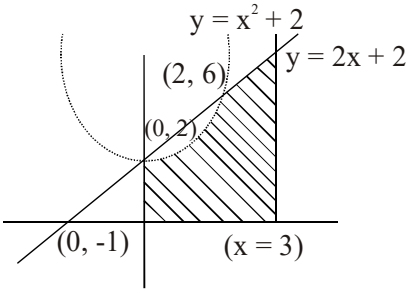
$\alpha = 169, \beta = 2$

35. Let the area of the region $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min \{x^2 + 2, 2x + 2\}\}$ be A. Then 12A is equal to :
[JEE(Main)Jan.2024]

माना क्षेत्र $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min \{x^2 + 2, 2x + 2\}\}$ का क्षेत्रफल A है। तो 12A बराबर है।
[JEE(Main)Jan.2024]

Ans. 164

Sol.



$$A = \int_0^2 \left(x^2 + 2 \right) dx + \int_2^3 (2x + 2) dx$$

$$A = \left(\frac{x^3}{3} + 2x \right)_0^2 + \left(x^2 + 2x \right)_2^3$$

$$A = \frac{41}{3} \Rightarrow 12A = 164$$

36. The area of the region enclosed by the parabola $(y - 2)^2 = x - 1$, the line $x - 2y + 4 = 0$ and the positive coordinate axes is _____.
[JEE(Main)Jan.2024]

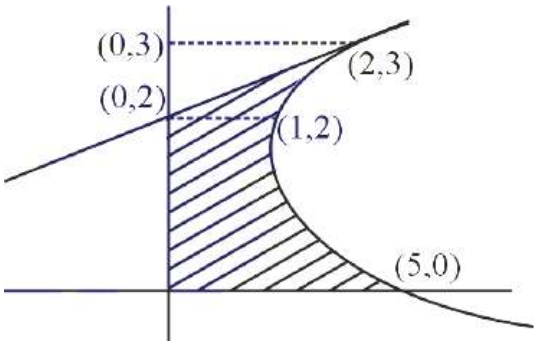
परवलय $(y - 2)^2 = x - 1$, रेखा $x - 2y + 4 = 0$ तथा धनात्मक निर्देशांक अक्षों से घिरे क्षेत्र का क्षेत्रफल _____
[JEE(Main)Jan.2024]

Ans. 5

Sol. Solving the equations

$(y - 2)^2 = x - 1$ and $x - 2y + 4 = 0$

$x = 2(y - 2)$



$\frac{x^2}{4} = x - 1$

$x^2 - 4x + 4 = 0$

$(x - 2)^2 = 0$

$x = 2$

Exclose area (w.r.t. y-axis) $= \int_0^3 x\,dy - \text{Area of } \Delta$

$= \int_0^3 \left((y-2)^2 + 1\right) dy - \frac{1}{2} \times 1 \times 2$

$= \int_0^3 \left(y^2 - 4y + 5\right) dy - 1$

$= \left[\frac{y^3}{3} - 2y^2 + 5y\right]_0^3 - 1$

$= 9 - 18 + 15 - 1 = 5$

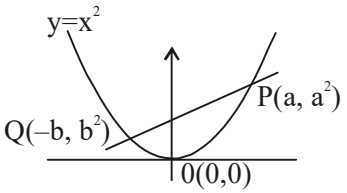
37. Three points O(0, 0), P(a, a^2), Q(–b, b^2), a > 0, b > 0, are on the parabola $y = x^2$. Let S_1 be the area of the region bounded by the line PQ and the parabola, and S_2 be the area of the triangle OPQ. If the minimum value of $\frac{S_1}{S_2}$ is $\frac{m}{n}$, gcd(m, n) = 1, then m + n is equal to _____. **[JEE(Main)Jan.2024]**

परवलय $y = x^2$ पर तीन बिंदु O(0, 0), P(a, a^2), Q(–b, b^2), a > 0, b > 0 हैं। माना रेखा PQ तथा परवलय से घिरे क्षेत्र का क्षेत्रफल S_1 है तथा त्रिभुज OPQ का क्षेत्रफल S_2 है। यदि $\frac{S_1}{S_2}$ का न्यूनतम मान $\frac{m}{n}$, gcd(m, n) = 1 हैं, तो m + n बराबर है _____

[JEE(Main)Jan.2024]

Ans. 7

Sol.



Equation of PQ

$$y+b=\frac{a^2-b^2}{a+b}\left(x-b^2\right)$$

$$\Rightarrow y=(a-b)x+ab$$

$$S_1=\int_{-b}^{a}\left((a-b)x+ab-x^2\right)dx$$

$$=\left(\frac{(a-b)}{2}x^2+abx-\frac{x^3}{3}\right)_{-b}^{a}$$

$$=\frac{1}{6}(a+b)^3$$

$$S_2=\frac{1}{2}\begin{vmatrix}-b & b^2 & 1\\ 0 & 0 & 1\\ a & a^2 & 1\end{vmatrix}=\frac{1}{2}ab(a+b)$$

$$\frac{S_1}{S_2}=\frac{\frac{1}{6}(a+b)^3}{\frac{1}{2}\cdot ab(a+b)}$$

$$=\frac{1}{3}\frac{(a+b)^2}{ab}=\frac{1}{3}\left(\frac{a}{b}+\frac{b}{a}+2\right)$$

$$\because \frac{b}{a}+\frac{a}{b}\geq 2$$

$$\Rightarrow\left(\frac{S_1}{S_2}\right)_{min}=\frac{4}{3}=\frac{m}{n}$$

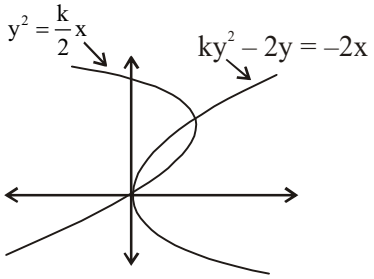
$$\Rightarrow m+n=7$$

38. The sum of squares of all possible values of k, for which area of the region bounded by the parabolas $2y^2 = kx$ and $ky^2 = 2(y - x)$ is maximum, is equal to _____. **[JEE(Main)Jan.2024]**

k के सभी संभव मानों, जिनके लिए परवलयों $2y^2 = kx$ तथा $ky^2 = 2(y - x)$ से घिरे क्षेत्र का क्षेत्रफल अधिकतम है, के वर्गों का योग है _____ **[JEE(Main)Jan.2024]**

Ans. 8

Sol.



$$\text{Area} = A = \int_0^{\frac{2k}{k^2+4}} \left[\left(\frac{ky^2 - 2y}{-2} \right) - \left(\frac{2y^2}{k} \right) \right] dy$$

$$\text{Area} = \left[\left(\frac{-k}{2} - \frac{2}{k} \right) \frac{y^3}{3} + \frac{y^2}{2} \right]_0^{\frac{2k}{k^2+4}}$$

$$\text{Area} = \frac{-\left(k^2+4\right)}{2k} \times \left(\frac{2k}{k^2+4} \right)^3 \times \frac{1}{3} + \left(\frac{2k}{k^2+4} \right)^2 \times \frac{1}{2}$$

$$= -\left(\frac{2k}{k^2+4} \right)^2 \times \frac{1}{3} + \left(\frac{2k}{k^2+4} \right)^2 \times \frac{1}{2}$$

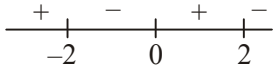
$$= \frac{1}{6} \times \frac{4k^2}{\left(k^2+4\right)^2}$$

$$A(k) = \frac{2}{3} \frac{k^2}{\left(k^2+4\right)^2}$$

$$A'(K) = \frac{2}{3} \frac{2K\left(K^2+4\right)^2 - 2K^2\left(K^2+4\right)(2K)\Big]}{\left(K^2+4\right)^4} = 0$$

$$\Rightarrow \frac{2}{3}(2K)\left(K^2+4\right)\left[K^2+4-2K^2\right] = 0$$

K = 0 or K = 2 or K = –2



$\therefore$ K = 0 is local minima

K = –2 and 2 are local maxima

$\therefore (-2)^2 + 2^2 = 8$

39. One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by the curves be $\frac{1}{24}\left(\ell\sqrt{5} + m\right) - n\log_e\left(1+\sqrt{5}\right)$, where $\ell, m, n \in N$. Then $\ell + m + n$ is equal to :

[JEE(Main)April,2024]

वक्रों $y = 1 + 3x - 2x^2$ तथा $y = \frac{1}{x}$ का एक प्रतिच्छेदन $\left(\frac{1}{2}, 2\right)$ है। माना इन वक्रों से घिरे का क्षेत्रफल $\frac{1}{24}\left(\ell\sqrt{5} + m\right) - n\log_e\left(1+\sqrt{5}\right)$ है, जहाँ $\ell, m, n \in N$ है, तो $\ell + m + n$ का मान होगा :

[JEE(Main)April,2024]

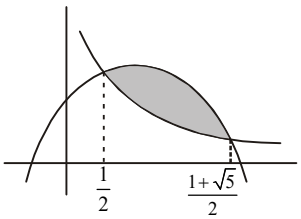
(A) 32 (B) 29 (C) 31 (D) 30

Ans. D

Sol. $2x^3 - 3x^2 - x + 1 = 0$

$\Rightarrow (2x - 1)(x^2 - x - 1) = 0$

$\Rightarrow x = \frac{1}{2}, x = \frac{1+\sqrt{5}}{2}$



$$\text{Area} = \int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}} \left(1 + 3x - 2x^2 - \frac{1}{x}\right)dx$$

$$= \left[x + + \frac{3x^2}{2} - \frac{2x^3}{3} - \ell nx\right]$$

$$= \frac{\sqrt{5}+1}{2} + \frac{3}{8}\left(\sqrt{5}+1\right)^2 - \frac{1}{12}\left(\sqrt{5}+1\right)^3 - \ell n\left(\frac{\sqrt{5}+1}{2}\right) - \left(\frac{1}{2} + \frac{3}{8} - \frac{1}{12} - \ell n\frac{1}{2}\right)$$

$$= \frac{1}{24}\left[12\left(\sqrt{5}+1\right) + 9\left(\sqrt{5}+1\right)^2 - 2\left(\sqrt{5}+1\right)^3 - 12 - 9 + 2\right] - \ell n\left(\frac{\sqrt{5}+1}{2} \times 2\right)$$

$$= \frac{1}{24}\left[12\left(\sqrt{5}+1\right) + 9\left(6 + 2\sqrt{5}\right) - 2\left(5\sqrt{5} + 1 + 3\sqrt{5}\left(\sqrt{5}+1\right) - 19\right)\right] - \ell n\left(\sqrt{5}+1\right)$$

$$= \frac{1}{24}\left[14\sqrt{5} + 15\right] - \ell n\left(\sqrt{5}+1\right)$$

$\therefore l = 14, m = 15, n = 1$

$l + m + n = 30$

40. The are (in sq. units) of the region described by $\{(x, y) : y^2 \le 2x, \text{ and } y \ge 4x - 1\}$ is :

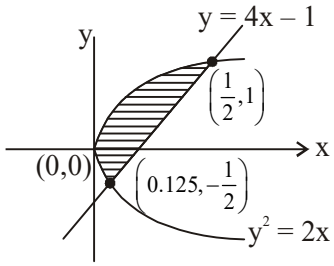
[JEE(Main)April,2024]

क्षेत्र $\{(x, y) : y^2 \le 2x$ तथा $y \ge 4x - 1\}$ का क्षेत्रफल (वर्ग इकाईयों में) है : **[JEE(Main)April,2024]**

(A) $\frac{8}{9}$ (B) $\frac{9}{32}$ (C) $\frac{11}{12}$ (D) $\frac{11}{32}$

Ans. B

Sol. Area $= \int_{-\frac{1}{2}}^{1}\left(\frac{y+1}{4}-\frac{y^2}{2}\right)dy$



$$=\left[\frac{y^2}{8}+\frac{y}{4}-\frac{y^3}{6}\right]_{-\frac{1}{2}}^{1}$$

$$=\left(\frac{1}{8}+\frac{1}{4}-\frac{1}{6}\right)-\left(\frac{1}{32}-\frac{1}{8}+\frac{1}{48}\right)$$

$$=\frac{5}{24}+\frac{7}{96}$$

$$=\frac{27}{96}$$

$$=\frac{9}{32}$$

41. The area enclosed between the curves $y = x|x|$ and $y = x - |x|$ is : **[JEE(Main)April,2024]**

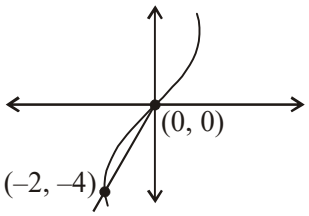
वक्रों $y = x|x|$ तथा $y = x - |x|$ के बीच घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) 2/3 (B) 4/3 (C) 8/3 (D) 1

Ans. B

Sol. $y = x|x|$ & $y = x - |x|$

$$y = x|x| = \begin{cases} x^2 & , x > 0 \\ -x^2 & , x < 0 \end{cases}$$



$$y = x - |x| = \begin{cases} 0 & , x > 0 \\ 2x & , x < 0 \end{cases}$$

point of intersections are (0 , 0) & (–2, 4).

$$\therefore \text{Area} = \int_{-2}^{0} \left(-x^2 - 2x\right) dx$$

$$= \left[\frac{-x^3}{3} - \frac{2x^2}{2}\right]_{-2}^{0}$$

$$= -\left[\frac{8}{3} - 4\right] = \frac{4}{3} \text{ sq. unit}$$

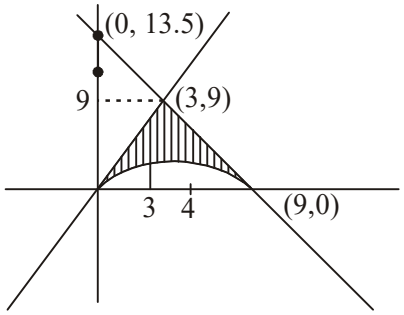
42. Let the area of the region enclosed by the curves $y = 3x$, $2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then 10A is equal to : **[JEE(Main)April,2024]**

माना वक्रों $y = 3x$, $2y = 27 - 3x$ और $y = 3x - x\sqrt{x}$ से घिरे क्षेत्र का क्षेत्रफल A है, तो 10A बराबर है : **[JEE(Main)April,2024]**

(A) 184 (B) 172 (C) 154 (D) 162

Ans. D

Sol.



$$y = 3x - x^{\frac{3}{2}} \qquad y' = 3 - \frac{3}{2}\sqrt{x}$$

$$y' = 3\left(1 - \frac{\sqrt{x}}{2}\right)$$

$f'(x)$ + –

Area of shaded region

$$= \int_0^3 \left(3x - \left(3x - x^{\frac{3}{2}}\right)\right) dx + \int_3^9 \frac{27 - 3x}{2} - \left(3x - x^{\frac{3}{2}}\right) dx$$

$$= \int_0^3 x^{\frac{3}{2}} dx + \frac{1}{2}\int_3^9 \left(27 - 9x + 2x^{\frac{3}{2}}\right) dx = \left(\frac{2}{5}x^{\frac{5}{2}}\right)_0^3 + \frac{1}{2}\left[27x - \frac{9x^2}{2} + 2\times\frac{2}{5}x^{\frac{5}{2}}\right]_3^9$$

$$= \frac{2}{5}\times 9\sqrt{3} + \frac{1}{2}\left[243 - \frac{729}{2} + \frac{4}{5}\times 81\times 3 - 81 + \frac{81}{2} - \frac{4}{5}\times 9\sqrt{3}\right]$$

$$= \frac{1}{2}\left[\frac{486 - 729 - 81}{2} + \frac{972}{5}\right] = \frac{81}{5}$$

$\Rightarrow 10A = 162$

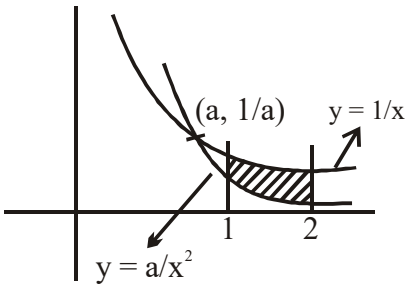
43. If the area of the region $\left\{(x, y) : \frac{a}{x^2} \le y \le \frac{1}{x}, 1 \le x \le 2, 0 < a < 1\right\}$ is $\left(\log_e 2\right) - \frac{1}{7}$ then the value of $7a - 3$ is equal to : **[JEE(Main)April,2024]**

यदि क्षेत्र $\left\{(x, y) : \frac{a}{x^2} \le y \le \frac{1}{x}, 1 \le x \le 2, 0 < a < 1\right\}$ का क्षेत्रफल $\left(\log_e 2\right) - \frac{1}{7}$ है, तो $7a - 3$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) 1 (B) 2 (C) 0 (D) –1

Ans. D

Sol. Required Area $= \int_1^2 \left(\frac{1}{x} - \frac{a}{x^2}\right) dx = \left(\ln x + \frac{a}{x^2}\right) = \ln 2 + \frac{a}{2} - a$



Now, $\ln 2 + \frac{a}{2} - a = \ln 2 - \frac{1}{7}$

$\frac{a}{2} = \frac{1}{7}$ $\Rightarrow$ $7a = 2$ and $7a - 3 = -1$

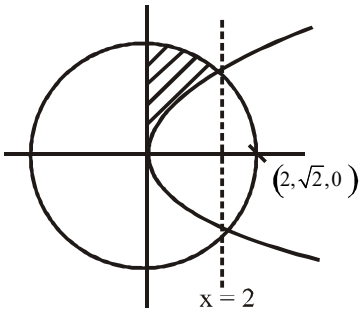
44. The area of the region in the first quadrant inside the circle $x^2 + y^2 = 8$ and outside the parabola $y^2 = 2x$ is equal to: **[JEE(Main)April,2024]**

प्रथम चतुर्थांश में वृत्त $x^2 + y^2 = 8$ के अन्दर तथा परवलय $y^2 = 2x$ के बाहर के क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) $\pi-\frac{2}{3}$ (B) $\pi-\frac{1}{3}$ (C) $\frac{\pi}{2}-\frac{1}{3}$ (D) $\frac{\pi}{2}-\frac{2}{3}$

Ans. A

Sol.



Requried Area = Ar(circle from 0 to 2) – ar(para from 0 to 2)

$$=\int_0^2\sqrt{8-x^2}dx-\int_0^2\sqrt{2x}dx$$

$$=\left[\frac{x}{2}\sqrt{8-x^2}+\frac{8}{2}\sin^{-1}\frac{x}{2\sqrt{2}}\right]_0^2-\sqrt{2}\left[\frac{x\sqrt{x}}{\frac{3}{2}}\right]_0^2$$

$$=\frac{2}{2}\sqrt{8-4}+\frac{8}{2}\sin^{-1}\frac{2}{2\sqrt{2}}-\frac{2\sqrt{2}}{3}\left(2\sqrt{2}-0\right)$$

$$\Rightarrow 2+4\cdot\frac{\pi}{4}-\frac{8}{3}=\pi-\frac{2}{3}$$

45. The parabola $y^2 = 4x$ divides the area of the circle $x^2 + y^2 = 5$ in two parts. The area of the smaller part is equal to:

[JEE(Main)April,2024]

परवलय $y^2 = 4x$, वृत्त $x^2 + y^2 = 5$ के क्षेत्रफल को दो भागों में बाँटता है। छोटे भाग का क्षेत्रफल है :

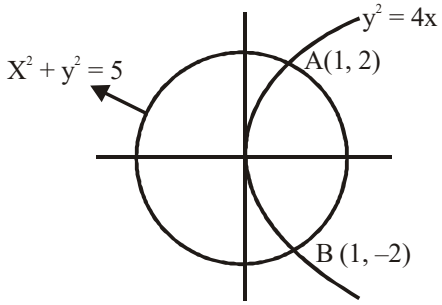
[JEE(Main)April,2024]

(A) $\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (B) $\frac{1}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(C) $\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (D) $\frac{2}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Ans. C

Sol.



$$\text{Area of smaller region} = \int_{-2}^{2}\left(\sqrt{5-y^2}-\frac{y^2}{4}\right)dy$$

$$= 2\int_{0}^{2}\left(\sqrt{5-y^2}-\frac{y^2}{4}\right)dy$$

$$= 2\left[\left(\frac{y}{2}\sqrt{5-y^2}+\frac{5}{2}\sin^{-1}\left(\frac{y}{\sqrt{5}}\right)\right)_0^2-\frac{1}{12}\left(y^3\right)_0^2\right]$$

$$= 2\left[1+\frac{5}{2}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)-\frac{2}{3}\right]=\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

46. The area of the region enclosed by the parabolas $y = x^2 - 5x$ and $y = 7x - x^2$ is _____.

[JEE(Main)April,2024]

परवलयों $y = x^2 - 5x$ तथा $y = 7x - x^2$ से घिरे क्षेत्र का क्षेत्रफल है _____ **[JEE(Main)April,2024]**

Ans. 198

Answer by Matrix is (72)

Sol. $y = x^2 - 5x$

$y = 7x - x^2$

Now $x^2 - 5x = 7x - x^2$

$x = 0, 6$

$$\text{So Area} = \int_0^6\left(x^2-5x-7x+x^2\right)dx$$

$$= 2\frac{x^3}{3}-12\frac{x^2}{2}\Big]_0^6$$

$$= \left[=2\times\frac{6^3}{3}-6\times36\right)0$$

$$= \frac{2\times6\times6\times6}{3} - 6\times36$$

$$= 36 \times 4 - 6 \times 36$$

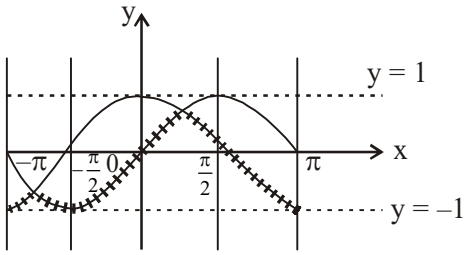
$$= 36(-2) = 72$$

47. Let the area of the region enclosed by the curve $y = \min\{\sin x, \cos x\}$ and the x-axis between $x = -\pi$ to $x = \pi$ be A. Then A^2 is equal to _____. **[JEE(Main)April,2024]**

वक्र $y = \min\{\sin x, \cos x\}$ तथा x-अक्ष के बीच परिबद्ध क्षेत्र का क्षेत्रफल A है। जहाँ x-अक्ष $x = -\pi$ तथा $x = \pi$ के बीच है, तो A^2 का मान होगा _____ **[JEE(Main)April,2024]**

Ans. 16

Sol. $y = f(x) = \min\{\sin x, \cos x\}$



$$A = -\int_{-\pi}^{\frac{-3\pi}{4}} \cos x dx - \int_{\frac{-3\pi}{4}}^{0} \sin x dx + \int_{0}^{\frac{\pi}{4}} \sin x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x dx$$

$$-\int_{\frac{\pi}{2}}^{\pi} \cos x dx$$

$A = 4$

$A^2 = 16$